

MicroTraitLLM: A RAG-based, Microbe-focused LLM for Researchers

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Abstract

Rapid advances in Large Language Models (LLMs) and broad potential applications in biological research make them a compelling point of investigation. Key challenges remain, including the tendency of LLMs to hallucinate if they are not provided with domain-specific information, thereby potentially misinforming users; the small number of LLM applications directed to prokaryotes despite significant advances in microbiome studies, especially when compared to the many human-specific and eukaryotic domain-specific LLM application that have recently been released; and the inability of current microbe-specific LLM tools to provide sufficiently comprehensive, accurate, and timely answers with proper citations.

We present MicroTraitLLM, a retrieval-augmented generation (RAG) LLM which utilizes zero-shot and single-shot prompting to give specific, citation-based answers for researchers. Its connection to the live-updating PubMed Central Open Access article database allows the tool to remain up-to-date on scientific knowledge. MicroTraitLLM flexibly allows the user to customize their experience by selecting their choice of LLMs. The tool is also able to generate accurate citations in various formats. The tool is also able to generate accurate citations in various formats. We present empirical results demonstrating that MicroTraitLLM provides informative responses as judged by microbial experts and LLM-as-a-judge, while not increasing latency time compared to popular commercial LLMs.

Keywords

Large Language Model, Artificial Intelligence, RAG, Microbes, PubMed Central, Natural Language Processing, LLM-as-a-judge, ChatGPT, Llama, Gemma,

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1 Introduction

Within the past several decades, single cell sequencing and metagenomics have revealed deeper insights into prokaryotes, especially around functional traits, providing a wide-ranging knowledge base for researchers[1]. These functional traits describe "measurable characteristics", such as how various microbial taxa survive in their environment of choice, as well as the underlying chemical functions that occur within said microbes [1][12]. However, the high throughput of prokaryote genomic data produces large content sizes that humans may struggle to manually process.

Large Language Models (LLMs) have recently emerged as tools for information retrieval, synthesis, and prediction in the bioinformatics and biomedical fields. Through Natural Language Processing (NLP), these LLMs are capable of processing vast amounts of written text information within a much shorter span than a human. LLMs have also been shown to provide a comprehensive summarization of the information consumed through the engineered prompts provided by the user [13].

However, there are drawbacks to using biological and biomedical domain-specific LLMs. *First*, many LLMs are trained up to a certain date, creating a "knowledge cutoff," a date after which the LLM lacks information about recent breakthroughs or developments. This limitation poses a significant challenge for users seeking comprehensive, up-to-date information retrieval. *Second*, available tools such as AskMicrobe.com only incorporate a single LLM for response and evaluation[7]. Users may have alternative preferences for LLM use based on previous experiences or lower costs for model inference. *Third*, LLMs are vulnerable to "hallucinations" that may cause them to provide incorrect information [6]. Many bioinformatics LLM tools tend to avoid aligning their models with returning potential hallucinations, which may limit the impact of such applications.

Here, we present MicroTraitLLM, an integrated LLM-tool that returns information on microbial traits by parsing academic articles in the PubMedCentral (PMC) Open Access and NCBI databases [8]. MicroTraitLLM takes a given question as input and returns a summary answer that includes citations and hyperlinks to additional



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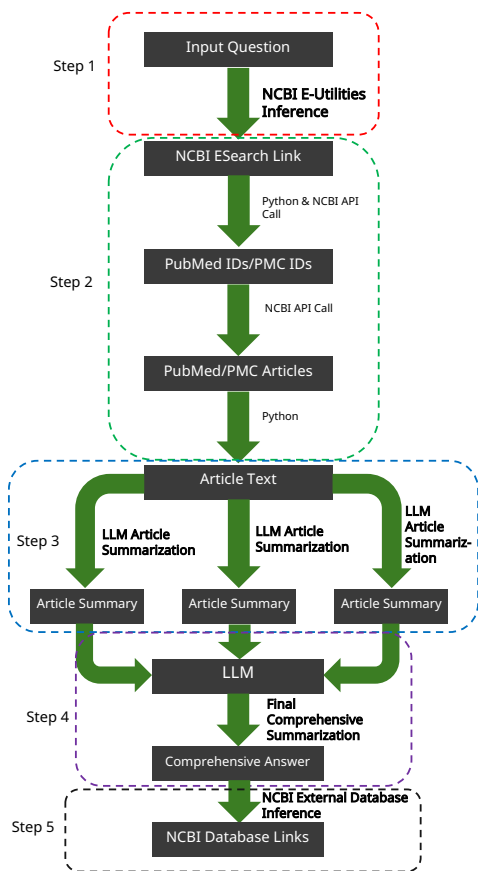


Figure 1: Visual overview of code structure in MicroTraitLLM. All LLM inference steps are bolded.

information that may interest the user, such as genes, proteins, and relevant taxonomies of interest. In doing so, MicroTraitLLM attempts to reduce the risk of hallucinations that could otherwise include problematic, and even dangerous misinformation in responses. Providing the citations and hyperlinks also offers transparency into its response and source material. MicroTraitLLM is designed to work with CPUs, thus not requiring intensive amounts of RAM or GPU, but also provides the option to utilize local GPU resources when available.

2 Methods

Figure 1 illustrates the steps taken to generate the full article text of the summary: 1) the algorithm is initiated by a user question; 2) the tool makes various NCBI E-utilities API calls; 3) parallel LLM processes creates several articles summaries; 4) a LLM makes a final summary from the individual article summaries; 5) a LLM identifies unique key terms, and autonomous code creates NCBI database links.

2.1 Code Structure

2.1.1 MicroTraitLLM Initiation. MicroTraitLLM initially takes a user question and an article parse number input (See top of Figure

1). Other optional parameters that a user can select are: (a) LLM selection, with the default model set as Llama3.2:3b, and (b) desired model temperature, with default temperature of 0.0.

2.1.2 NCBI E-Utilities API. The selected LLM is prompted to create the term portion of an NCBI ESearch API link based on the user input and single-shot prompting (See Section 2.2 for additional information). The search term is incorporated into a NCBI ESearch URL (See Step 2 in Figure 1), which returns various PubMedCentral identifications (PMcIDs). The PMcIDs are used to form BioC base URLs, which return article text and metadata in a large JSON library format (See Step 4 in Figure 1)[4]. The article text is filtered so that it is devoid of graphs, tables, and other forms of media that may confuse LLMs in later steps (See fifth box in Figure 1).

2.1.3 LLM Article Summarization. MicroTraitLLM will attempt to open as many instances of child processes as the user requested to review, with a fallback to opening as many child processes as possible in case of hardware limitation. Each child process is given its own instance of the user-selected LLM so there is no influence from previous articles when creating the individual summaries. Each child process LLM receives one article text and metadata to form its summary (indicated in Step 3 in Figure 1). The child process returns a confidence score for relevance of the article to the user question alongside an abridged version of the article. The confidence score is used in the following section to prioritize more relevant articles over others. Citations are autonomously created during each child process.

2.1.4 Final Comprehensive Summarization. Once all articles are summarized, the list of summaries is provided to an isolated instance of the selected LLM. The prompt for this instance of the LLM can be seen in Section 2.2. The LLM is tasked with synthesizing an answer for the user based on the article summaries (indicated in Step 4 in Figure 1).

2.1.5 NCBI External Database Link Creation. The system creates a set of external database links with relevant information after answer generation. These links redirect the user to the NCBI taxonomy, protein, and gene databases. An individual instance of LLM is prompted to parse through the summary and identify any taxa, proteins, or genes mentioned in the response.

When creating links to the NCBI taxonomy database, the NCBI taxonomy ID (taxID) is first identified by creating a NCBI E-utilities API URL for the taxa in the prompt. The relevant NCBI taxIDs are then used to create URLs to the relevant webpages. When creating links to the NCBI protein and gene databases, both the NCBI TaxID and the protein/gene IDs must be identified so no irrelevant but similarly named protein or gene terms are returned. All URLs are returned to the graphical user interface as additional information for the user.

2.2 Prompt Engineering

2.2.1 Prompting Overview. MicroTraitLLM makes use of zero and single shot prompting to inform the LLM of its specific task. As outlined in Fig. 1, the LLM is prompted 4 times: 1) initially creating the link for accessing the NCBI E-Utilities API; 2) creating individual summaries for each article; 3) creating a full summary

from the summaries of the articles; 4) searching the generated answer for details regarding taxonomy, proteins, or genes. Prompts were designed through a trial and error process when using the ChatGPT-4o-mini, Gemma2:2b, and Llama3.2:3b models with the objective of creating an accurate response to trial questions at each stage of the process. Each system prompt is followed by a single-shot example, which provides the structure for an ideal output. During the prompt designing process, it was most beneficial to be direct about the desired output from the tested models, including a single-shot answer which appeared to reduce the amount of variability in response. Furthermore, models appeared to provide better responses when they were tasked with a small, precise task. Additionally, according to White et al. (2024), adding a personality to an LLM prompt can generate a more comprehensive answer [11], which is reflected in the prompting below. In all following prompt engineering statements, the "citation format" term is the user-selected citation format.

2.2.2 Initial Prompt. Before initial URL creation in Step 1, MicroTraitLLM is provided the following single-shot prompt in order to generate the search term:

You are tasked with generating a single search term filter for NCBI API URLs based on a user question. The search term filter consists of various important key words that are spaced apart by plus signs. An example question is: 'What bacterial strains can degrade TNT?'. The appropriate filter term for the example question is 'bacterial+strains+degrade+TNT'.

2.2.3 Article Summarization Prompt. In the multiprocessing of individual articles, an article summary is generated using the following initial system prompt:

You are an expert in researching microbial metagenomics and microbial traits; however, you have no prior knowledge of any information or topic involving microbes. You are tasked with summarizing a provided article in the context of a given question from the user. All information required to answer the question will be provided to you in a large text format. Your answer must be 5–8 sentences answering the user question with detailed insights based on what you read in the article. Additionally, when writing your response, you must cite all sources that are provided to you in {citation format} format. You must also provide a grade after the {citation format} citation for each article. The grade should be on a scale of 0–100, with the grade corresponding to how much the article assisted in answering the user question. You must answer the user question to the best of your ability without using any other information besides the information provided to you. Question: What are some bacterial strains associated with ear infections?

MicroTraitLLM is also provided a single shot prompt (See GitHub in Section 7).

2.2.4 Comprehensive Summary Prompt. The system prompt in full summary generation is as follows:

You are an expert in microbial metagenomics and microbial traits. You are tasked with answering the question provided by the user. All information required to answer the question will be provided to you in a large text format. Your answer should be a detailed paragraph consisting of 5–10 sentences answering the user question. Additionally, when writing your response, you must cite every source that you use, and every source should be cited in {citation format} format. You must have both in-text citations and a references section. If you cannot cite the given information in {citation format} format, please instead list the article title and its corresponding ID. You should prioritize using the articles with a higher grade over the ones with a lower grade. You must answer the user question to the best of your ability without using any other information besides the information provided to you.

MicroTraitLLM is also provided a single shot prompt (See GitHub in Section 7).

2.3 Database

MicroTraitLLM utilizes the PMC Open Access Subset [8]. The dataset consists of millions of publicly-available journal articles, providing a large and authoritatively reliable resource for microbe-adjacent articles. This specific database fulfills two desired requirements: up-to-date information parsing, ensuring the tool's longevity, as well as open dataset access such that any user can utilize MicroTraitLLM effectively. Furthermore, the NCBI taxonomy, gene, and protein databases operate within a similar system, allowing for efficient supplementary data parsing. Together with the PMC Open Access Subset, these data sources provide state-of-the-art knowledge for MicroTraitLLM.

2.4 Models

MicroTraitLLM can utilize ChatGPT-4o-mini, Meta Llama 3.2:3b, Google Gemma2:2b, or Llama3.3:70b-versatile as the LLM for its process. Each of the models was selected due to their small parameter size or external server inference, as well as their capacity for large information summarization. Llama3.2:3b and Gemma2:2B are imported into the tool through Ollama [9]. Ollama allows MicroTraitLLM to utilize CPUs for inferences from low-parameter LLMs, thereby allowing computers with weaker hardware to still utilize local model inference. The tool can utilize GPUs if they are available; otherwise, CPUs are utilized to run the models locally. ChatGPT-4o-mini utilizes the OpenAI API to access OpenAI servers and models for inference, while Llama3.3:70b-versatile uses GroqCloud as its main server for inference.

2.5 Example Input/Output

Figure 2 shows a sample input and output to our tool, MicroTraitLLM. The user first asks a question in the microbial domain. The user also selects the number of articles for the LLM to read through and summarize, the preferred LLM to use, and the citation format that they would like to have provided in the answer. For this example, the number of summary articles was 8, the LLM was Llama3.3:70b-versatile, and the citation format was APA.

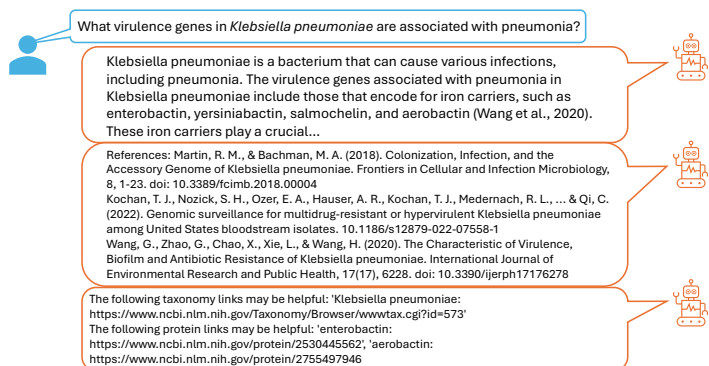


Figure 2: Visual overview of example input and output use of MicroTraitLLM.

After providing the question and processing, MicroTraitLLM then provides a response to the user question as the first output. This initial response contains in-text citations and detailed analysis from relevant articles. The second segment of MicroTraitLLM's response is the references section, where all articles are cited in the format of choice for the user. This provides transparency to the user so they can review the selected articles at their discretion. The third segment of the response contains the set of links to external databases. Each of these links are then presented to the user alongside their respective key term so the user can further investigate the related terminology mentioned in the answer.

2.6 Evaluation Technique Overviews

2.6.1 Computational Resource Evaluation Protocol. Each model's responses were evaluated based on GPU usage (reported in GPU hours), Memory Usage, and runtime per single instance of running the model. Since average CPU usage was minimal, it was omitted from computational resource testing. The desired time to response for MicroTraitLLM is to match popular LLMs, i.e., approximately 15-20 seconds. Only local models were subjected to computational resource evaluation, as any model that accesses external servers for LLM inference has little impact on standard resource allocation.

2.6.2 LLM-as-a-judge Protocol. Utilizing LLMs as judges has become a significant point of research in recent years. As LLMs have become more powerful, they have outstripped traditional metrics such as BLEU and METEOR, which rely on surface-level token overlap and fail to capture the coherence and context-awareness of modern LLMs [2][5]. One alternative is LLM-as-a-judge, where LLMs themselves function as evaluators, as flexible prompting and increased training allows for improved response and evaluation of less tangible metrics [3]. Here, we use LLM-as-a-judge to evaluate the quality of response for microbial trait related answers through a Likert scale ranging from zero to ten. Zero represents a completely irrelevant answer and 10 being an exceptional answer in this scale. Justifications were also requested as part of the prompt. Each evaluations set consists of $n=99$ samples; however, due to the nature of LLM response, the occasional LLM-as-a-judge answer did not return a Likert scale answer, causing imbalanced sample sizes. The LLM used for evaluation was DeepSeek-R1, specifically selected due

to its Chain-of-Thought response training. Additionally, Wataoaka et al. (2024) demonstrate LLMs tend to prefer responses from similar models[10]. We therefore utilize DeepSeek to avoid affinity bias between models.

2.6.3 Human Evaluation Protocol. Human participants were asked to evaluate a set of microbial trait questions and LLM responses. The participants were blinded to the model that generated the answer. Each participant was assigned 5 random questions, each of which had a MicroTraitLLM answer, and an alternative model answer. Each set of answers had identical parameters. Each human participant was asked to evaluate the response on a Likert scale from 0-10, with 0 being considered a completely irrelevant answer, and 10 being an ideal answer. Answer groups were established based on what LLM was used. Each surveyor reviewed 10 answers per survey, with 5 for one randomized model and 5 being for another randomized model.

2.6.4 Metrics. Quantitative evaluations were run based on Large Language Model. We measure computational resource usage for locally run models, as well as time to response completion for comparison against commercial LLMs. Furthermore, we provide human evaluations of experts comparing MicroTraitLLM to commercial LLMs.

3 Results

3.1 LLM Computational Resource Evaluation

Local models Gemma2:2b and Llama3.2:3b were subjected to a sample of 100 successive tests to test computational resource usage. Overall, Gemma2:2b required less GPU usage (64%) and GPU time (455 minutes) compared to Llama3.2:3b (77% and 641 minutes, respectively). GPU memory consumption for both models remained relatively stable at approximately 3900 MB. CPU memory usage varied between 6450-6550 MB for Llama3.2 and 6600-6800 MB for Gemma2.

All LLMs were subject to a time to response analysis, shown in Figure 3. On average, Llama3.3:70b-versatile was the fastest model, with an average response time of 8.4 seconds. The local models of llama3.2:3b and Gemma2:2b performed similarly to ChatGPT-4o-mini, which uses the OpenAI servers for LLM inference. The commercial LLM ChatGPT provided second-fastest average answer at around 10 seconds, and AskMicrobe provided the longest average answer of around 24 seconds.

A one-way ANOVA was conducted to determine significant differences between groups. Once statistical significance was determined between at least one of the groups (p -value: $1.239e-154$), the Tukey's HSD Pairwise comparisons test was conducted, which concluded that Llama3.2:3b and Gemma2:2b respond within a statistically comparable time frame. However, all other models respond at statistically different times from one another.

3.2 LLM-as-a-judge Evaluation

LLM-as-a-judge was used to evaluate MicroTraitLLM and commercial LLM responses. All answers were graded on a Likert scale from zero to ten, with zero representing an irrelevant score and ten being a superior answer. In the case where a grade was not provided,

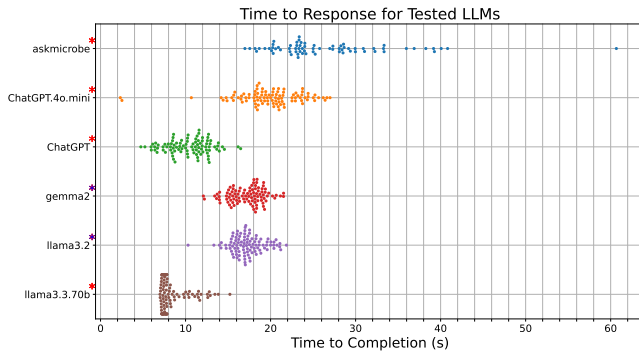


Figure 3: Swarmplot for time to response for Tested LLMs. Commerical LLMs (AskMicrobe and ChatGPT-base) were timed individually via their website. Red asterisks indicate statistical significance ($p < 0.05$). Llama3.2:3b and Gemma2:2b have purple asterisks, because they were similar to one another but significantly different from all other groups.

the score was omitted as an NaN value. The Kruskal-Wallis non-parametric test was used to test for significant differences between the values. Dunn’s Test with a Bonferroni adjustment was used as the post-hoc test.

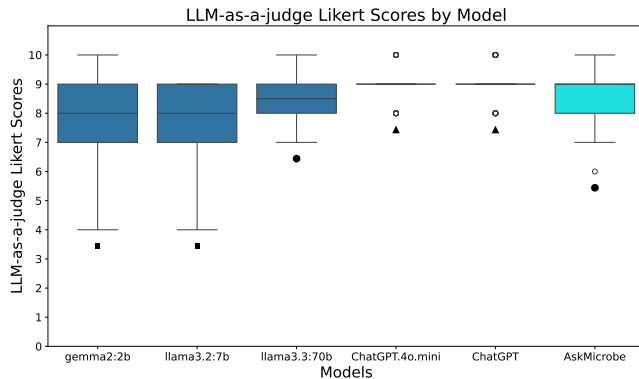


Figure 4: LLM-as-a-judge scores for various models utilized in MicroTraitLLM (dark blue) and comparison models (light blue). Each filled-in shape indicates a group that performs statistically identical to each other but different from other groups. For example, gemma2:2b and llama3.2:3b performed almost identically, and therefore have a similar marking.

The mean LLM score was approximately 8-9 for all models (see Figure 4). The smaller local models in Gemma2: 2b ($n = 92$, mean: 7.94, SD: 0.95) and Llama3.2:3b ($n = 96$, mean: 7.91, SD: 1.05) had the highest variation in score. Meanwhile, the stronger model, such as Llama3.3:70b-versatile ($n=81$, mean: 8.44, SD: 0.68) showed a bit more consistency in its scoring. Additionally, the commercial LLM AskMicrobe.com ($n=81$, mean: 8.58, SD: 0.66) also remained consistent. Interestingly, the base ChatGPT and ChatGPT-4o-mini scored identically, with $n = 58$, a mean of 9.01, and a standard deviation of 0.57.

The Kruskal-Wallis test determined that there was at least one statistically significant difference between the groups (p -value=0.00189). Therefore, Dunn’s test with a Bonferroni correction revealed three distinct groups of similar performance: Gemma2:2b & Llama3.2:3b as one group, AskMicrobe.com & Llama3.3:70 as the second, and the two ChatGPT models as the final. These three groups can be identified in Figure 4. However, there are two other statistically non-significant groups that also appear: AskMicrobe.com and the ChatGPT models ($p = 0.135$ when comparing AskMicrobe against either ChatGPT model), and the local models plus Llama3.3:70b-versatile forming a second correlated group ($p = 0.463$ between Gemma2:2b and Llama3.3:70b-versatile, $p = 0.391$ between Llama3.2:3b and Llama 3.3:70b). These findings suggest a potential bias in DeepSeek-R1 towards ChatGPT-like responses, possibly due to alignment in training data or stylistic similarity. Furthermore, there is statistical evidence that the ChatGPT models tended to have the best response according to DeepSeek, followed by AskMicrobe.com and Llama3.3:70b-versatile, with the local models performing the worst.

3.3 Expert Answer Evaluation

7 human surveyors were given various LLM responses and asked to grade them. All answers were graded on a Likert scale from zero to ten, with zero representing an irrelevant score and ten being a superior answer. In the case where a grade was not provided by the LLM, the score was omitted as an NaN value.

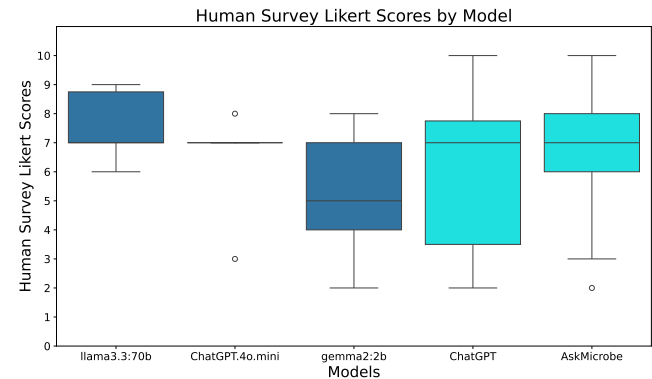


Figure 5: Boxplot of human evaluator survey, where $n = 7$ surveyors were tested. Dark blue models (left) indicate models that are used by MicroTraitLLM, whereas light blue models (right) indicate commercially available models. There were too few samples to assess statistical significance.

Overall, the surveyors on average showed preference towards Llama3.3:70b-versatile, with a mean score of 7.5. Other models had a wider range of scores, with AskMicrobe and ChatGPT-4o-mini obtaining mean scores of 6.44 and 6.4 respectively. Gemma2:2b was the worst performing model, with a mean score of 5.25.

4 Discussion

MicroTraitLLM can provide an answer in a eight to fifteen second window, which is competitive with commercial LLMs. AskMicrobe.com on average requires 24 seconds to respond, slower than

both MicroTraitLLM and standard commercial ChatGPT. The time to response for MicroTraitLLM is heavily dependent on the model. The local models in Gemma2:2b and Llama3.2:3b are dependent on memory size and GPU strength; therefore, their similar performance in response time may be due to the hardware specifications of the trial PC. Furthermore, the local models use significantly more computational resources, but CPU and GPU memory appears fairly consistent across most metrics. If the selected model is ChatGPT, the system becomes restricted by the OpenAI server queue.

The LLM-as-a-judge evaluations reveal that Gemma2:2b and Llama3.2:3b still perform worse on average in the LLM-as-a-judge standard, even after specific prompting and direct literature application. Furthermore, there may be some level of bias within the DeepSeek model towards ChatGPT model answers, as they were scored exactly the same across 106 total different answers while receiving the highest scores in the test. Wataoka et al. (2024) note that LLM-as-a-judge tends to prefer answers of lower perplexity [10]. The ChatGPT models tend to provide lower perplexity answers than other models, which might cause the potential bias seen here.

However, the results offer some potential insights into the expert preferences in the field. Despite limits in sample size, the results shown here demonstrate that human experts tend to prefer answers provided by stronger, newer models. Furthermore, there appears to still be necessary improvement in provided answers.

5 Limitations

This work contains several limitations. Firstly, the system prompting could be improved, providing more in-depth results than presented here. Our human evaluation was limited to seven microbial experts, which restricts the generalization of our findings regarding answer quality preferences. Notable technical limitations include: 1) MicroTraitLLM's reliance on the PMC Open Access database, which excludes alternative publications that may contain critical information, thereby risking incomplete or biased responses for certain queries. 2) The computational resource evaluation was conducted on a single hardware configuration, though performance may vary across systems.

The LLM-as-a-judge evaluation findings suggest that DeepSeek-R1 may have a preference for ChatGPT-style responses, potentially due to similarities in training approaches or response formatting. Furthermore, our evaluation focused primarily on response quality and computational efficiency, but did not assess critical factors such as factual accuracy of the generated summaries, consistency of citations, or the tool's ability to handle edge cases (e.g., queries with no relevant literature, contradictory findings across papers, or highly specialized terminology poorly represented in the literature). Finally, while MicroTraitLLM offers flexibility in model selection, we did not evaluate the consistency of responses across different models for identical queries, which could impact user trust and research reproducibility.

6 Future Works

Future work will investigate further refined prompt engineering, such as by using optimization methodologies such as evolutionary methods to achieve stronger prompt design. Another area of future

work includes additional assessments of more powerful models and their impact. One of the strengths of the MicroTraitLLM tool is that it is designed to be LLM-agnostic, allowing it to keep up with the continuous advances in open-source and proprietary frontier LLMs. Furthermore, the addition of more databases could lead to even more innovative answers, providing a stepping stone for further research. NCBI, while containing a comprehensive database, does not contain all relevant publications and information regarding various topics. Additional databases would likely improve MicroTraitLLM expert response. Finally, future development builds may add a stage where sources can be evaluated for reliability or provide an indication in the response of how authoritative they may be. This could be based, for example, on the number of citations, journal reputation, analysis of the rigor in the article, etc.

7 Code Repository

Code for MicroTraitLLM can be found at the following GitHub: <https://github.com/EESI/MicroTraitLLM>

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